Genome-wide Association Analysis - Data Quality Control

Introduction

In this exercise, you will learn how to perform data quality control (QC) by removing markers and samples that fail QC quality control criteria. You will also examine your samples for individuals that are related to each other and/or are duplicate samples. Each sample will also be tested for excess homozygosity and heterozygosity of genotype data. Each SNP will be tested for deviations from Hardy-Weinberg Equilibrium. These exercises will be carried out using PLINK 1.9.

In this exercise, there is a separate handout with questions and fill-in tables. You will be instructed to fill-in the tables and figures throughout this exercise, which can be filled in using the associated handout.

1. Using PLINK

PLINK can upload data in different formats please see the PLINK documentation (https://www.cog-genomics.org/plink/1.9/input) for additional details. The data for this exercise is in PLINK/LINKAGE file format. There are two files: a pedfile (`GWAS.ped`) and a map file (`GWAS.map`).Please note the commands must be given in the directory where the data reside. For this exercise the directory which contains the files for the exercise is `data` , which you can navigate on the file directory on the left sidebar.

Please examine these files and the PLINK documentation. To preview the files run the commands below.

In [1]:
```
# Set working directory
cd ./data
```

In [2]:
```
plink --file GWAS
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
(C) 2005-2024 Shaun Purcell, Christopher Chang   GNU General Public License
v3
Logging to plink.log.
Options in effect:
  --file GWAS

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6424 variants, 248 people).
--file: plink.bed + plink.bim + plink.fam written.
```

Note, that PLINK outputs a file called plink.log that contains the same output which you see on the screen. To see all options, type `plink --help` for more information. Determine how many samples there are in your data set and fill in Oval 1 of the flowchart on the handout.

2. Data Quality Control

a. Removing Samples and SNPs with Missing Genotypes.

You will exclude samples that are missing more than 10% of their genotype calls. These samples are likely to have been generated using low quality DNA and can also have higher than average genotyping error rates.

In [3]:
```
plink --file GWAS --mind 0.10 --recode --out GWAS_clean_mind
```

```
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v3
Logging to GWAS_clean_mind.log.
Options in effect:
  --file GWAS
  --mind 0.10
  --out GWAS_clean_mind
  --recode

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6424 variants, 248 people).
--file: GWAS_clean_mind-temporary.bed + GWAS_clean_mind-temporary.bim +
GWAS_clean_mind-temporary.fam written.
6424 variants loaded from .bim file.
248 people (125 males, 123 females) loaded from .fam.
1 person removed due to missing genotype data (--mind).
ID written to GWAS_clean_mind.irem .
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see GWAS_clean_mind.hh ); many
commands treat these as missing.
Total genotyping rate in remaining samples is 0.996863.
6424 variants and 247 people pass filters and QC.
Note: No phenotypes present.
--recode ped to GWAS_clean_mind.ped + GWAS_clean_mind.map ... done.
```

Examine `GWAS_clean_mind.log` to see how many samples are excluded based on this criterion and fill in Box 1.

We will now create two versions of your dataset, one with SNPs with a minor allele frequencies (MAFs) >5% and the other with SNPs with a MAFs <5%.

You will now remove SNPs with MAFs>5% that are missing >5% of their genotypes and then remove SNPs with MAFs<5% that are missing >1% of their genotypes. SNPs which are missing genotypes can have higher error rates than those SNP markers without missing data.

In [4]:
```
plink --file GWAS_clean_mind --maf 0.05 --recode --out MAF_greater_5
plink --file GWAS_clean_mind --exclude MAF_greater_5.map --recode --out MAF_
plink --file MAF_greater_5 --geno 0.05 --recode --out MAF_greater_5_clean
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
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v3
Logging to MAF_greater_5.log.
Options in effect:
  --file GWAS_clean_mind
  --maf 0.05
  --out MAF_greater_5
  --recode

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6424 variants, 247 people).
--file: MAF_greater_5-temporary.bed + MAF_greater_5-temporary.bim +
MAF_greater_5-temporary.fam written.
6424 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see MAF_greater_5.hh ); many
commands treat these as missing.
Total genotyping rate is 0.996863.
556 variants removed due to minor allele threshold(s)
(--maf/--max-maf/--mac/--max-mac).
5868 variants and 247 people pass filters and QC.
Note: No phenotypes present.
--recode ped to MAF_greater_5.ped + MAF_greater_5.map ... done.
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
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v3
Logging to MAF_less_5.log.
Options in effect:
  --exclude MAF_greater_5.map
  --file GWAS_clean_mind
  --out MAF_less_5
  --recode

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6424 variants, 247 people).
--file: MAF_less_5-temporary.bed + MAF_less_5-temporary.bim +
MAF_less_5-temporary.fam written.
6424 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
--exclude: 556 variants remaining.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Total genotyping rate is 0.996913.
556 variants and 247 people pass filters and QC.
Note: No phenotypes present.
--recode ped to MAF_less_5.ped + MAF_less_5.map ... done.
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
```

```
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Logging to MAF_greater_5_clean.log.
Options in effect:
  --file MAF_greater_5
  --geno 0.05
  --out MAF_greater_5_clean
  --recode

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (5868 variants, 247 people).
--file: MAF_greater_5_clean-temporary.bed + MAF_greater_5_clean-temporary.bi
m +
MAF_greater_5_clean-temporary.fam written.
5868 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see MAF_greater_5_clean.hh ); man
y
commands treat these as missing.
Total genotyping rate is 0.996858.
2 variants removed due to missing genotype data (--geno).
5866 variants and 247 people pass filters and QC.
Note: No phenotypes present.
--recode ped to MAF_greater_5_clean.ped + MAF_greater_5_clean.map ... done.
```

Fill in Box 2a.

In [5]:
```
plink --file MAF_less_5 --geno 0.01 --recode --out MAF_less_5_clean
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
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v3
Logging to MAF_less_5_clean.log.
Options in effect:
  --file MAF_less_5
  --geno 0.01
  --out MAF_less_5_clean
  --recode

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (556 variants, 247 people).
--file: MAF_less_5_clean-temporary.bed + MAF_less_5_clean-temporary.bim +
MAF_less_5_clean-temporary.fam written.
556 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Total genotyping rate is 0.996913.
59 variants removed due to missing genotype data (--geno).
497 variants and 247 people pass filters and QC.
Note: No phenotypes present.
--recode ped to MAF_less_5_clean.ped + MAF_less_5_clean.map ... done.
```

Fill in Box 2b. Merge the two files.

In [6]:
```
plink --file MAF_greater_5_clean --merge MAF_less_5_clean.ped MAF_less_5_cle
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)            cog-genomics.org/plink/1.
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v3
Logging to GWAS_MAF_clean.log.
Options in effect:
  --file MAF_greater_5_clean
  --merge MAF_less_5_clean.ped MAF_less_5_clean.map
  --out GWAS_MAF_clean
  --recode

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (5866 variants, 247 people).
--file: GWAS_MAF_clean-temporary.bed + GWAS_MAF_clean-temporary.bim +
GWAS_MAF_clean-temporary.fam written.
247 people loaded from GWAS_MAF_clean-temporary.fam.
247 people to be merged from MAF_less_5_clean.ped.
Of these, 0 are new, while 247 are present in the base dataset.
5866 markers loaded from GWAS_MAF_clean-temporary.bim.
497 markers to be merged from MAF_less_5_clean.map.
Of these, 497 are new, while 0 are present in the base dataset.
Performing single-pass merge (247 people, 6363 variants).
Merged fileset written to GWAS_MAF_clean.bed + GWAS_MAF_clean.bim +
GWAS_MAF_clean.fam .
6363 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see GWAS_MAF_clean.hh ); many
commands treat these as missing.
Total genotyping rate is 0.99716.
6363 variants and 247 people pass filters and QC.
Note: No phenotypes present.
--recode ped to GWAS_MAF_clean.ped + GWAS_MAF_clean.map ... done.
```

A more stringent criterion for missing data is used, samples missing >3% of their genotypes are removed.

In [7]:
```
plink --file GWAS_MAF_clean --mind 0.03 --recode --out GWAS_clean2
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
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v3
Logging to GWAS_clean2.log.
Options in effect:
  --file GWAS_MAF_clean
  --mind 0.03
  --out GWAS_clean2
  --recode

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6363 variants, 247 people).
--file: GWAS_clean2-temporary.bed + GWAS_clean2-temporary.bim +
GWAS_clean2-temporary.fam written.
6363 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
0 people removed due to missing genotype data (--mind).
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see GWAS_clean2.hh ); many comman
ds
treat these as missing.
Total genotyping rate is 0.99716.
6363 variants and 247 people pass filters and QC.
Note: No phenotypes present.
--recode ped to GWAS_clean2.ped + GWAS_clean2.map ... done.
```

Fill in Box 3.

b. Checking Sex

Error of the reported sex of an individual can occur. Information from the SNP genotypes can be used to verify the sex of individuals, by examining homozygosity (F) on the X chromosome for every individual. F is expected to be <0.2 in females and >0.8 in males. To check sex run

In [8]:
```
plink --file GWAS_clean2 --check-sex --out GWAS_sex_checking
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)             cog-genomics.org/plink/1.
9/
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v3
Logging to GWAS_sex_checking.log.
Options in effect:
  --check-sex
  --file GWAS_clean2
  --out GWAS_sex_checking

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6363 variants, 247 people).
--file: GWAS_sex_checking-temporary.bed + GWAS_sex_checking-temporary.bim +
GWAS_sex_checking-temporary.fam written.
6363 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see GWAS_sex_checking.hh ); many
commands treat these as missing.
Total genotyping rate is 0.99716.
6363 variants and 247 people pass filters and QC.
Note: No phenotypes present.
--check-sex: 186 Xchr and 0 Ychr variant(s) scanned, 5 problems detected.
Report written to GWAS_sex_checking.sexcheck .
```

Examine the `GWAS_sex_checking.sexcheck` file at the command line and determine if there are individuals whose recorded sex is inconsistent with genetic sex.

In [9]:
```
awk 'NR==1 || $5=="PROBLEM"' GWAS_sex_checking.sexcheck
```

```
     FID       IID       PEDSEX       SNPSEX       STATUS            F
NA20506   NA20506            2            1      PROBLEM            1
NA20530   NA20530            2            1      PROBLEM            1
NA20757   NA20757            2            0      PROBLEM       0.2141
NA20766   NA20766            2            0      PROBLEM       0.2292
NA20771   NA20771            2            0      PROBLEM       0.2234
```

NA20530 and NA20506 were coded as a female (2) and from the genotypes appear to be males (1). In addition, 3 individuals (NA20766, NA20771 and NA20757) do not have enough information to determine if they are males or females and PLINK reports sex = 0 for the genotyped sex. Fill in the information for table 1:

Reasons for these kinds of discrepancies, include the records are incorrect, incorrect data entry, sample swap, unreported Turner or Klinefelter syndromes. Additionally, if a sufficient number of SNPs have not been genotyped on the X chromosome it can be difficult to accurately predict the sex of an individual. In this dataset, there are only 194 X chromosomal SNPs. If you cannot validate the sex of the individual they should be removed. For this exercise, we are going to assume that when the sex was checked, we found it was incorrectly recorded (i.e. these samples were male). Therefore, this error could simply be corrected.

c. Duplicate Samples

The following PLINK command can be used to check for duplicate samples

In [10]:
```
plink --file GWAS_clean2 --genome --out duplicates
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
(C) 2005-2024 Shaun Purcell, Christopher Chang   GNU General Public License
v3
Logging to duplicates.log.
Options in effect:
  --file GWAS_clean2
  --genome
  --out duplicates

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6363 variants, 247 people).
--file: duplicates-temporary.bed + duplicates-temporary.bim +
duplicates-temporary.fam written.
6363 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
Using up to 9 threads (change this with --threads).
Before main variant filters, 247 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see duplicates.hh ); many command
s
treat these as missing.
Total genotyping rate is 0.99716.
6363 variants and 247 people pass filters and QC.
Note: No phenotypes present.
Excluding 186 variants on non-autosomes from IBD calculation.
IBD calculations complete.
Finished writing duplicates.genome .
```

Examine the `duplicates.genome` file at the command line:

In [11]:
```
head duplicates.genome
```

```
   FID1      IID1    FID2      IID2 RT     EZ       Z0       Z1       Z2   PI_HAT PH
E        DST      PPC    RATIO
   1328   NA06984    1328   NA06989 OT      0   1.0000   0.0000   0.0000   0.0000   -
1  0.733459  0.0773   1.8238
   1328   NA06984    1330   NA12340 UN     NA   1.0000   0.0000   0.0000   0.0000   -
1  0.737882  0.3859   1.9617
   1328   NA06984    1330   NA12342 UN     NA   1.0000   0.0000   0.0000   0.0000   -
1  0.728677  0.0030   1.6720
   1328   NA06984    1330   NA12343 UN     NA   0.9615   0.0278   0.0108   0.0246   -
1  0.741146  0.8064   2.1196
   1328   NA06984    1334   NA12144 UN     NA   1.0000   0.0000   0.0000   0.0000   -
1  0.740049  0.1046   1.8453
   1328   NA06984    1334   NA12145 UN     NA   0.9298   0.0702   0.0000   0.0351   -
1  0.740988  0.9080   2.1875
   1328   NA06984    1334   NA12146 UN     NA   0.9874   0.0072   0.0054   0.0090   -
1  0.737671  0.5089   2.0030
   1328   NA06984    1334   NA12239 UN     NA   0.9807   0.0193   0.0000   0.0097   -
1  0.735452  0.7559   2.0960
   1328   NA06984    1340   NA06994 UN     NA   0.9642   0.0244   0.0113   0.0236   -
1  0.740967  0.7809   2.1067
```

We are interested in the Pi-Hat (the estimated proportion IBD sharing) value. You may notice that there is more than one duplicate (Pi-Hat=~1). Also, examine the output for pairs of individuals with high Pi-Hat values which can indicate they are related. The amount of allele sharing [Z(0), Z(1) and Z(2)] across all SNPs provides information on the type of relative pair.

In [12]:
```
awk 'NR==1 || $10>0.4' duplicates.genome
```

```
   FID1      IID1    FID2      IID2 RT     EZ       Z0       Z1       Z2   PI_HAT PH
E        DST      PPC    RATIO
   1344   NA12057   13291   NA25001 UN     NA   0.0000   0.0025   0.9975   0.9988   -
1  0.999594  1.0000       NA
   1444   NA12739    1444   NA12748 OT      0   0.0026   0.9949   0.0025   0.5000   -
1  0.835832  1.0000  867.0000
   1444   NA12739    1444   NA12749 OT      0   0.0026   0.9807   0.0168   0.5071   -
1  0.838175  1.0000  869.0000
   M033   NA19774    M041   NA25000 UN     NA   0.0000   0.0000   1.0000   1.0000   -
1  1.000000  1.0000       NA
```

Note: Pi-hat can be inflated and individuals appear to be related to each other if you have samples from different populations. This explains why we observe pairs of individuals with Pi-hat >0.05 since three distinct populations were analyzed. Additionally, this phenomenon can be observed if a subset(s) of samples have higher genotyping/sequencing error rates, which creates two or more "populations" and the individuals within these "populations" incorrectly appear to be related.

Using the command below, observe how many sample pairs have pi-hat >0.05:

In [13]:
```
awk 'NR==1 {print $1, $2, $3, $4, $10} $10>0.05 {print $1, $2, $3, $4, $10}'
```

FID1 IID1 FID2 IID2 PI_HAT
FID1 IID1 FID2 IID2 PI_HAT
1328 NA06984 1347 NA11882 0.0677
1328 NA06984 1451 NA12776 0.0546
1328 NA06989 1341 NA06993 0.0537
1328 NA06989 1451 NA12776 0.0501
1328 NA06989 1459 NA12875 0.0525
1330 NA12340 1341 NA06993 0.0671
1330 NA12340 1353 NA12489 0.0616
1330 NA12340 1423 NA11918 0.0530
1330 NA12340 1444 NA12750 0.0740
1330 NA12340 1458 NA12842 0.0536
1330 NA12340 1463 NA12889 0.0543
1330 NA12340 NA20506 NA20506 0.0533
1330 NA12340 NA20530 NA20530 0.0518
1330 NA12343 1340 NA07056 0.0551
1330 NA12343 1341 NA06985 0.0658
1330 NA12343 1341 NA06993 0.0537
1330 NA12343 1346 NA12044 0.0606
1330 NA12343 1347 NA11882 0.0866
1330 NA12343 1350 NA11829 0.0577
1330 NA12343 1350 NA11831 0.0555
1330 NA12343 1353 NA12546 0.0765
1330 NA12343 1358 NA12716 0.0568
1330 NA12343 1362 NA11992 0.0585
1330 NA12343 1362 NA11995 0.0541
1330 NA12343 1377 NA11892 0.0578
1330 NA12343 1408 NA12156 0.0580
1330 NA12343 1418 NA12273 0.0691
1330 NA12343 1421 NA12282 0.0613
1330 NA12343 1421 NA12287 0.0616
1330 NA12343 1423 NA11918 0.0520
1330 NA12343 1444 NA12739 0.0795
1330 NA12343 1444 NA12749 0.0597
1330 NA12343 1444 NA12750 0.0712
1330 NA12343 1447 NA12762 0.0652
1330 NA12343 1458 NA12842 0.0508
1330 NA12343 1458 NA12843 0.0512
1330 NA12343 1459 NA12873 0.0662
1330 NA12343 13281 NA12347 0.0504
1330 NA12343 NA20506 NA20506 0.0545
1330 NA12343 NA20515 NA20515 0.0673
1330 NA12343 NA20522 NA20522 0.0631
1330 NA12343 NA20539 NA20539 0.0833
1330 NA12343 NA20543 NA20543 0.0604
1330 NA12343 NA20544 NA20544 0.0526
1330 NA12343 NA20581 NA20581 0.0521
1330 NA12343 NA20586 NA20586 0.0568
1330 NA12343 NA20766 NA20766 0.1091
1330 NA12343 NA20769 NA20769 0.0556
1330 NA12343 NA20805 NA20805 0.0635
1330 NA12343 NA20810 NA20810 0.0723
1334 NA12144 1334 NA12146 0.0526
1334 NA12144 1375 NA12264 0.0510
1334 NA12144 1444 NA12750 0.0550
1334 NA12144 NA20544 NA20544 0.0700

1334 NA12144 NA20796 NA20796 0.0623
1334 NA12145 1334 NA12146 0.0547
1334 NA12145 1340 NA07000 0.0878
1334 NA12145 1341 NA06985 0.0501
1334 NA12145 1345 NA07345 0.0587
1334 NA12145 1347 NA11882 0.0734
1334 NA12145 1362 NA11994 0.0604
1334 NA12145 1408 NA12156 0.0554
1334 NA12145 1418 NA12273 0.0697
1334 NA12145 1421 NA12282 0.0875
1334 NA12145 1423 NA11918 0.0511
1334 NA12145 1423 NA11920 0.0511
1334 NA12145 NA20515 NA20515 0.0626
1334 NA12145 NA20518 NA20518 0.0516
1334 NA12145 NA20544 NA20544 0.0635
1334 NA12146 1340 NA06994 0.0587
1334 NA12146 1340 NA07000 0.0631
1334 NA12146 1341 NA06993 0.0667
1334 NA12146 1344 NA12057 0.0740
1334 NA12146 1345 NA07346 0.0507
1334 NA12146 1347 NA11882 0.0504
1334 NA12146 1349 NA11839 0.0736
1334 NA12146 1349 NA11840 0.0559
1334 NA12146 1353 NA12489 0.0592
1334 NA12146 1354 NA12400 0.0580
1334 NA12146 1362 NA11992 0.0618
1334 NA12146 1362 NA11993 0.0741
1334 NA12146 1362 NA11995 0.0555
1334 NA12146 1375 NA12264 0.0542
1334 NA12146 1377 NA11893 0.0667
1334 NA12146 1408 NA12156 0.0641
1334 NA12146 1418 NA12272 0.0588
1334 NA12146 1418 NA12273 0.0600
1334 NA12146 1418 NA12275 0.0518
1334 NA12146 1423 NA11920 0.0607
1334 NA12146 1444 NA12750 0.0581
1334 NA12146 1447 NA12760 0.0543
1334 NA12146 1447 NA12761 0.0536
1334 NA12146 1451 NA12776 0.0707
1334 NA12146 1451 NA12778 0.0603
1334 NA12146 1459 NA12875 0.0597
1334 NA12146 1463 NA12892 0.0538
1334 NA12146 13281 NA12347 0.0504
1334 NA12146 13291 NA25001 0.0732
1334 NA12146 13292 NA07031 0.0583
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NA20801	NA20801	NA20802	NA20802	0.0838

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NA20806 NA20806 NA20818 NA20818 0.0561
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NA20818 NA20818 NA20828 NA20828 0.0816
```

Create the following txt file:

```
1344 NA12057
1444 NA12739
M033 NA19774
```

name it `IBS_excluded.txt` and save it to the folder with your PLINK data. Give the command:

In [14]:
```
cat > IBS_excluded.txt << 'EOF'
1344 NA12057
1444 NA12739
M033 NA19774
EOF
```

In [15]:
```
plink --file GWAS_clean2 --remove IBS_excluded.txt --recode --out GWAS_clear
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)            cog-genomics.org/plink/1.
9/
(C) 2005-2024 Shaun Purcell, Christopher Chang   GNU General Public License
v3
Logging to GWAS_clean3.log.
Options in effect:
  --file GWAS_clean2
  --out GWAS_clean3
  --recode
  --remove IBS_excluded.txt

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6363 variants, 247 people).
--file: GWAS_clean3-temporary.bed + GWAS_clean3-temporary.bim +
GWAS_clean3-temporary.fam written.
6363 variants loaded from .bim file.
247 people (125 males, 122 females) loaded from .fam.
--remove: 244 people remaining.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see GWAS_clean3.hh ); many comman
ds
treat these as missing.
Total genotyping rate in remaining samples is 0.997225.
6363 variants and 244 people pass filters and QC.
Note: No phenotypes present.
--recode ped to GWAS_clean3.ped + GWAS_clean3.map ... done.
```

Fill in Box 4 and Oval 3.

As part of QC usually the data is examined for outliers by plotting the first and second principal or multidimensional scaling (MDS) components. Using a subset of markers that have been pruned to remove LD (e.g., r2<0.1). Principal components analysis (PCA) and MDS will be performed in the second part of the exercise to detect outliers and control for populations substructure. Outlier can be due to study subjects coming from different populations e.g. European- and African-Americans or batch effects. If it is suspected that outliers are due to study subjects having been sampled from different populations than data from HapMap can be included to elucidate population membership, e.g. for a study of European-Americans if African-American study subjects are included they would cluster between the European and African HapMap samples. If you perform this type of analysis you should remove the HapMap samples and re-estimate the MDS or PC components before adjusting for population substructure or stratification. For this exercise data is used from HapMap Phase III which consists of CEU (Europeans from Utah), MEX (Mexicans from Los Angeles) and TSI (Tuscans from Italy). Three clusters can be observed that consist of the three data sets but no extreme outliers are observed. This data set is being used for demonstration purposes. Different populations should be analyzed separately and the results can be combined using meta-analysis. In part two of this exercise MDS and PC components will be constructed and analyzed.

d. Hardy-Weinberg Equilibrium (HWE):

To test for HWE we will test separately in each ancestry group and by case-control status. Therefore, we will need to use information on ancestry and cases-control status. Please note that this should be tested in the 3 different populations separately (CEU, MEX, TSI), but due to the small sample sizes, we tested it in the 3 populations together for example purposes. It should also be noted if the sample sizes are small it is difficult to detect a deviation from HWE.

In [16]:
```
plink --file GWAS_clean3 --pheno pheno.txt --pheno-name Aff --hardy
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
(C) 2005-2024 Shaun Purcell, Christopher Chang   GNU General Public License
v3
Logging to plink.log.
Options in effect:
  --file GWAS_clean3
  --hardy
  --pheno pheno.txt
  --pheno-name Aff

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6363 variants, 244 people).
--file: plink-temporary.bed + plink-temporary.bim + plink-temporary.fam
written.
6363 variants loaded from .bim file.
244 people (123 males, 121 females) loaded from .fam.
244 phenotype values present after --pheno.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see plink.hh ); many commands tre
at
these as missing.
Total genotyping rate is 0.997225.
--hardy: Writing Hardy-Weinberg report (founders only) to plink.hwe ... don
e.
```

Examine the file `plink.hwe` at the command line and look for SNPs with p-values of 10^{-7} or smaller.

In [17]:
```
awk 'NR==1 || $9<0.0000009' plink.hwe
```

```
 CHR                            SNP     TEST   A1   A2                GENO
O(HET)   E(HET)            P
   1                      rs2968487      ALL    T    C             0/124/114
0.521   0.3853    3.877e-10
   1                      rs2968487    UNAFF    T    C               0/83/70
0.5425   0.3953    2.262e-07
   1                     rs11239985      ALL    G    A             0/111/128
0.4644   0.3566    7.625e-08
   2                     rs12988076      ALL    C    A              69/82/93
0.3361   0.4952    6.973e-07
```

Using a criterion of $p < 10^{-7}$ to reject the null hypothesis of HWE, how many SNPs fail HWE in the cases? Fill out Oval 5 and Box 4. Using the same criteria, how many SNPs fail HWE in the controls? Complete Table 3 with this information.

Create a text file called `HWE_out.txt` with the following SNP in it:

```
rs2968487
```

and type the following command:

In [18]:
```
cat > HWE_out.txt << 'EOF'
rs2968487
EOF
```

In [19]:
```
plink --file GWAS_clean3 --exclude HWE_out.txt --recode --out GWAS_clean4
```

```
PLINK v1.9.0-b.8 64-bit (22 Oct 2024)              cog-genomics.org/plink/1.
9/
(C) 2005-2024 Shaun Purcell, Christopher Chang   GNU General Public License
v3
Logging to GWAS_clean4.log.
Options in effect:
  --exclude HWE_out.txt
  --file GWAS_clean3
  --out GWAS_clean4
  --recode

7836 MB RAM detected; reserving 3918 MB for main workspace.
.ped scan complete (for binary autoconversion).
Performing single-pass .bed write (6363 variants, 244 people).
--file: GWAS_clean4-temporary.bed + GWAS_clean4-temporary.bim +
GWAS_clean4-temporary.fam written.
6363 variants loaded from .bim file.
244 people (123 males, 121 females) loaded from .fam.
--exclude: 6362 variants remaining.
Using 1 thread (no multithreaded calculations invoked).
Before main variant filters, 244 founders and 0 nonfounders present.
Calculating allele frequencies... done.
Warning: 6 het. haploid genotypes present (see GWAS_clean4.hh ); many comman
ds
treat these as missing.
Total genotyping rate is 0.997229.
6362 variants and 244 people pass filters and QC.
Note: No phenotypes present.
--recode ped to GWAS_clean4.ped + GWAS_clean4.map ... done.
```

There are a number of SNPs with HWE p-values in the range of 10^{-5} to 10^{-6} in the controls. Based on above criterion they will not be excluded however, if they reach genome-wide significance during association testing they SNPs should be further investigated to ensure there is no genotyping error. You can now fill in Box 5 and Oval 4.